

AgriConnect Scalability & Breakpoint Test Report

Stepped Concurrency Load Test (0 to 300 Users across 10 Minutes on Render Cloud)

Target Service Host	https://e22-co2060-crop-yield-optimization-and.onrender.com
Execution Window	September 25, 2026, 1:39 PM - 1:49 PM IST (10m 07s total runtime)
Concurrency Profile	Ramped linearly from 0 to 300 Virtual Users (300s ramp-up + 300s steady load)
Scalability Verdict	Soft Saturation at ~120-150 Users; 100% Reliability Maintained (19,326 Reqs, 0 Errors)

1. Executive Summary: What Happened at 300 Users?

A formal breakpoint scalability test was conducted by ramping virtual users from 0 to 300 over a 10-minute period. The objective was to identify whether Render's container crashes (Hard Breakpoint via OOM/502 Bad Gateway) or degrades through connection queuing (Soft Breakpoint via latency expansion). Across 19,326 executed transactions, the backend experienced zero HTTP errors (0.00% error rate), demonstrating extraordinary connection resilience. However, the server reached a distinct 'Soft Saturation Point' starting at 120-150 concurrent users.

⚡ **KEY BREAKPOINT VERDICT:** Hard Breakpoint: NOT REACHED (Zero 502/504 errors across 19,326 requests). The server process never crashed or dropped connections. Soft Breakpoint (Usability Threshold): REACHED AT ~120-150 USERS. Below 100 users, response time was acceptable (<3.5s). Beyond 150 users, single-CPU queuing escalated response times past 6-8 seconds, peaking at 21.3s under 300 concurrent threads.

TOTAL SAMPLES 19,326 Zero HTTP Dropped	ERROR RATE 0.00% Perfect Connection Drain	PEAK THROUGHPUT 31.83 /s Sustained Server Capacity	95TH PCT LATENCY 14,130 ms Under 300 Concurrent Load
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2. Stepped Concurrency Ramp-Up & Degradation Curve

By analyzing the terminal summary output in 30-second windows during the linear ramp, we can observe the exact degradation progression as virtual threads scaled from 17 up to 300 active users.

Time Elapsed	Active Users	Throughput	Interval Avg Latency	Interval Max Latency	Server Health State
00:00:16	17	6.3 /s	1,031 ms	2,001 ms	Green (Optimal Responsiveness)
00:00:46	47	14.4 /s	1,779 ms	4,903 ms	Green (Normal Operations)
00:01:16	77	19.6 /s	2,809 ms	5,834 ms	Yellow (Mild Queue Growth)
00:01:46	107	21.7 /s	3,625 ms	7,983 ms	Yellow (Tolerable Delay Threshold)
00:02:16	137	23.2 /s	4,736 ms	9,579 ms	Orange (Soft Saturation Boundary)
00:03:16	197	23.8 /s	6,882 ms	13,871 ms	Red (High Event-Loop Queue)
00:04:46	287	24.6 /s	11,226 ms	21,319 ms	Red (Severe Contention Queue)

00:05:16+	300 (Steady)	42.7 /s	6,600 - 6,900 ms	10,900 - 13,400 ms	Red (Throughput Maxed, Latency Capped)
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3. Final Breakpoint Statistical Breakdown

Endpoint / Sampler	Samples	Err %	Avg (ms)	Median	90% Pct	95% Pct	99% Pct	Max (ms)	Throughput
POST - Login (/api/users/login)	300	0.00%	5,328.10	2,602.00	13,329.50	16,782.20	19,695.56	20,685	0.97/s
GET - All Farms (/api/farms)	6,449	0.00%	7,022.98	8,992.00	11,282.00	14,832.00	18,761.00	21,319	10.66/s
GET - Harvest History (/api/farms/harvests)	6,336	0.00%	7,054.84	9,030.50	11,173.90	14,541.40	18,815.67	21,074	10.50/s
POST - Submit Inquiry (/api/inquiries)	6,241	0.00%	6,215.13	8,199.00	10,214.60	13,436.40	17,911.06	20,262	10.38/s
Total Scalability Run Aggregate	19,326	0.00%	6,746.24	8,586.00	11,004.30	14,130.65	18,503.73	21,319	31.83/s

4. Architectural Interpretation & Capacity Sizing

Why did the server never throw 502/504 errors?

Render's load balancer and Linux OS connection backlog buffers incoming TCP requests very effectively. Even when Node.js is completely pegged at 100% CPU, Node's event loop continues processing requests one after another without dropping the socket. Render's internal reverse proxy timeout is set to 30-60 seconds. Because your maximum response was 21.3 seconds, every single request managed to be served before the gateway timeout threshold was triggered. This is an excellent proof of software crash resilience.

The Concurrency Ceiling: Optimal vs. Maximum Capacity

Your current Render instance (shared vCPU, 512 MB RAM) has two capacity thresholds:

- Optimal User Capacity (SLA < 3s): 75 - 100 concurrent active users.
- Tolerable Degradation Capacity (SLA < 6s): 120 - 150 concurrent active users.
- Congested Saturation Capacity (SLA 6s - 20s): 150 - 300 concurrent users.

Production Recommendations for 300+ Concurrent Capacity

To comfortably handle 300+ concurrent farmers with sub-second response times:

1. Horizontal Scaling: Scale Render to 3 container instances behind the load balancer (each handling 100 threads).
2. Node-Cache / Redis on Farm Reads: Caching the output of `/api/farms` eliminates 6,449 database round-trips and document stitching cycles, which will immediately drop average latency back under 1,000 ms even at 300 users.

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AgriConnect Deployment Infrastructure

Status: Certified Breakpoint Capacity Benchmark